

The Most Beautiful Room in the World. Creative Video VR/AR Experience Based on the Wedding Chamber Installation

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ABSTRACT

This creative video explains visually the development process of an interactive VR/AR immersive installation that reenacts a paramount masterpiece from the early Renaissance. “The Wedding Chamber Installation” is based on the aesthetic, artistic and pre-cinematic expressivity conveyed by the illusionistic devices created and painted by Andrea Mantegna in the late 15th century in his fresco “The Wedding Chamber”, dubbed as “The Most Beautiful Room in the World” situated in the Saint Giorgio Castle, Mantua, Italy.

INTRODUCTION

The “Most Beautiful Room in the World” video unfolds subsequently the theoretical and functional outcomes of the installation as a research study about how users and viewers perceive and manipulate the illusionistic component of the fresco through haptic responsive 3D models and animations. Andrea Mantegna’s influence over the Renaissance’s naturalistic rendition of the space is fundamental, and his works continued to inspire in the next centuries different schools of painting in Europe. His use of the concept of “Trompe L’oeil,” [1] inaugurates a proto-technological system that expands the architectural space of the fresco’s technique to illusionistic dimensions. However, recent studies have discovered that some of his frescos depict a more complex representation inspired by writers from the classical world, seeking to immerse the viewers in a more literary narrative and visual perceptions [2]. Those masterpieces followed an ambitious desire to play with illusion as a device to expand their pictorial narrative beyond their formal physical boundaries [3]. That is the case expressed in the series of frescos located at the Saint Giorgio Castle in The Palazzo Ducale di Mantua.

THE INSTALLATION EMPHASIS AND MOTIVATIONS

The immersive installation concept required an important amount of research, planning, development, and implementation solely based on the technological platforms needed to generate the visual content that ultimately conveys the implicit discourse, narrative, or message, articulated around a conceptual architecture that ponders the virtual and the physical space as performative interactive scenarios [4]. However, it is important to distinguish in the installation its content as the primary subject of research to achieve the construction and

articulation of the visual developments along with the technological resources. There is a strong research component associated with the historical, aesthetic, and artistic background of the subject of study around the Gonzaga family Weltanschauung deployed in the original frescos’ narrative. Moreover, there is the presence of Andrea Mantegna’s artistic mastery as a subject of study inconspicuously alluding the literary piece *The Dome* of Lucian of Samosata, as a classical influence from antiquity in terms of how this classic satirist ekphrasis method excerpts [5] influenced the manner Andrea Mantegna playfully articulates secular, and humanistic overtones in a time when religious commissions were the expression of patronage, and cultural expressions of society [6].

THE INSTALLATION TECHNICAL ASPECTS

The platform for development utilized was VRTK and MRTK in Unity 3D. Also, custom C# objects, and nodes for data collection were integrated. The 3D model and 3D animation developments were made in Maya software, and the advanced character’s motions in Autodesk Motion Builder. The XR component was developed using Visual Studio, MRTK, and Microsoft HoloLens. The main VR system utilized for visualization was HTC VIVE pro, HTC VIVE Plus, Oculus Quest 1, and Oculus Quest 2. The installation consisted in a central play area surrounded by three hanging panels altogether including a target image framed as a simulated ceiling to perform the oculus augmentation when is detected by the Mixed Reality HoloLens device.

CONCLUSION

The installation was presented at several international and regional venues. The effectiveness of interactions and responsiveness of its operative interface depends heavily on the viewer’s feedback and insights. For this reason, the associated study of user experiences has become a critical component in terms of a seminal analysis of data. The follow-up observation of the audience’s usage indicates two important variables that surfaced during the installation playing mode that directly impacts in how the viewer responds, the novelty of teleporting navigation, and the hand’s manipulation of interactive objects through haptics [7]. These two variables are persistently demonstrating that an immersive installation that implements XR (extended reality) necessarily requires a physical space to unfold and track the user’s body performance seeking the lively experience of “presence” in an immersive, or in an extended spatiality, where holographical 3D content is overlapped on top of physical objects.

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Andres Montenegro is an Associate Professor of Computer Animation at Purdue University Fort Wayne. U.S., He has participated in 2013 in the International Animation Festival of Annecy, France. In this event he has obtained a residency award in Paris at the DoubleMetre Animation studios for his short, animated film *The Little Quest of Petrovsky*. His Last animation film, *The Alley* has been screened in several international film festivals among them is the International Film Festival Canton, Guangzhou China. His immersive environment, XR (VR, AR, MR) installation *The Wedding Chamber* was presented at the ARTECH 2021: 10th International Conference on Digital and Interactive Arts October 2021 in Aveiro, Portugal, and the 2022 6th CAGA Conference of Expanded Animation, at the Hochschule University in Luzern Switzerland. His research work has been published in the Handbook of Research on Urban Tourism, Viral Society, and the Impact of the COVID-19 Pandemic, the International Journal of Art, Culture, Design, and Technology (IJACDT), and the 10th International Conference on Digital and Interactive Arts proceedings, ACM publication.

Audrey Ushenko has been a professor at Purdue University Fort Wayne since 1988. She received her B.A. from Indiana University-Bloomington in 1965, her M.A. in painting in 1967 from Northwestern University, and a Ph. D in Art History from Northwestern University in 1979. Dr. Ushenko teaches figure drawing, advanced painting, and advanced art history classes within the Fine Arts program. She has a very active painting career on a national level. She is represented by Denise Bibro Fine Art Inc. in New York City. Her goal is to bring out the best technical skill of students whether it is in drawing or painting while challenging the student to understand the historical context in which they are working.

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